

SOLUTIONS - Exercise - Week 9: MATH1061/7861

- (1) (4 marks) Let $A := \{5k + 2 \mid k \in \mathbb{Z}\}$. Does A have the same cardinality as $\mathbb{Z}$? Justify your answer.

The set A has the same cardinality as $\mathbb{Z}$. To show this, we will prove that the function $f : \mathbb{Z} \rightarrow A$ defined by $f(n) = 5n + 2$ is a bijection.

To show that f is surjective, take any $a \in A$. Then by the definition of A , $a = 5k + 2$ for some $k \in \mathbb{Z}$. Therefore $f(k) = 5k + 2 = a$, and so f is surjective.

To show that f is injective, suppose that $f(n) = f(m)$ for some $n, m \in \mathbb{Z}$. This means that $5n + 2 = 5m + 2$, which implies that $5n = 5m$ and hence $n = m$. Therefore f is injective.

As f is both surjective and injective, it is a bijection, and hence A has the same cardinality as $\mathbb{Z}$.

MARKING:

- 1 mark for identifying that they have the same cardinality
- 1 mark for constructing a valid bijection
- 1 mark for proving their construction is surjective
- 1 mark for proving their construction is injective

- (2) **(6 marks)** Let ρ be a relation on $\mathbb{Z}$ defined by $x\rho y$ if $a + b$ is even. Determine whether ρ is reflexive, symmetric, or transitive.

Reflexivity: Take any $n \in \mathbb{Z}$. Then $n + n = 2n$ is even, so $n\rho n$. Therefore ρ is reflexive.

Symmetry: Take any $n, m \in \mathbb{Z}$ and suppose that $n\rho m$. This means that $n + m$ is even. Because $n + m = m + n$, this implies that $m + n$ is also even, so $m\rho n$ and therefore ρ is symmetric.

Transitivity: Take any $n, m, k \in \mathbb{Z}$ and suppose that $n\rho m$ and $m\rho k$. This means that $n + m$ and $m + k$ are even, so $n + m = 2a$ and $m + k = 2b$ for some $a, b \in \mathbb{Z}$. Notice that

$$n + k = (n + m) + (k + m) - 2m = 2a + 2b - 2m = 2(a + b - m),$$

which implies that $n + k$ is even, and hence $n\rho k$. Therefore ρ is transitive.

MARKING:

For each part, 1 mark for correctly identifying that ρ has that property

For each part, 1 mark for providing a valid proof